World Models in Action

3. Representation learning before generation

This chapter covers

- The assumptions built into self-supervised objectives
- Contrastive, bootstrap, self-distillation and redundancy-reduction objectives
- Masked prediction and the choice of prediction target
- Latent bottlenecks and the limits of representation evaluation

A world model needs a representation that preserves distinctions relevant to predicting future observations. Pixels contain both useful information and details that a predictive model may not need. An encoder maps the observation x to a representation $z = f_\theta(x)$, but compression alone does not make z useful. Two observations that should lead to different predictions must not be merged merely because their images are similar.

When labels are unavailable, the training signal must be constructed from the data. Methods may identify a matching view among candidates, predict a representation produced by another branch, constrain feature statistics across a batch, or predict content hidden from the encoder. These objectives differ in their targets, their assumptions and their optimization dynamics.

Methods are attributed to their original papers, and paper-reported findings are distinguished from results independently reproduced for this book. The short code excerpt is attributed to the official PyTorch implementation; it illustrates the stated calculation rather than presenting a new training implementation.

The central question is therefore not just whether the training loss decreases. It is what the loss rewards the encoder for preserving. A feature may solve the pretraining task yet omit information needed for a later prediction. The following sections derive the principal objectives and explain exactly what their mathematical constraints do and do not establish.

3.1. Self-supervised learning foundations

The central distinction

Self-supervised learning constructs a relationship from data and optimizes an encoder to satisfy it. That relationship may be useful without being sufficient for a world model. Separate the training target, the learned representation and evidence about downstream state quality.

3.1.1. A representation is not just a compressed observation

Let x be an observation and let f_θ be an encoder with parameters θ. Its output is a vector z:

$$z = f_\theta(x).$$

A smaller representation is useful only if the information removed is unnecessary for the task. A vector may preserve object identity while losing location or orientation. It may describe a frame's appearance while losing motion that is apparent only from a sequence. Compression can remove nuisance detail, but it can also merge observations whose futures differ.

Chapter 2 formalized the desired contract in terms of sufficiency, minimality and aliasing. Self-supervised objectives do not observe that contract directly. They rely on a proxy relationship in the data. A representation may perform well on the proxy task while failing to retain information the world model needs.

3.1.2. Invariance is a decision

Suppose T_1 and T_2 are random transformations applied to the same input:

$$x_1 = T_1(x), \qquad x_2 = T_2(x).$$

A view-consistency objective encourages $f_\theta(x_1)$ and $f_\theta(x_2)$ to agree. The design decision is hidden in T_1 and T_2: they define which input differences the model is encouraged to ignore.

Random crops are useful when the cropped-out region is irrelevant to the intended representation. They are harmful when the missing region contains information needed to distinguish the scene or predict its next state. Color jitter can suppress nuisance lighting changes, but it can also remove a visual cue on which a later prediction depends. The transformation is part of the learning problem, not a neutral preprocessing step.

An ideal invariance would identify observations that the downstream task truly treats as equivalent. A practical augmentation only approximates that relation. We should not assume that two views are interchangeable for every task just because they came from the same image.

The distinction can be written directly into the objective. Let T_1, T_2 be random transformations sampled from a chosen distribution and let d measure disagreement between two features. A consistency objective has the form

$$L_{\text{cons}}(\theta) = \mathbb{E}_{x,\, T_1,\, T_2}\big[d\big(f_\theta(T_1 x),\, f_\theta(T_2 x)\big)\big].$$

The expectation averages over the views the augmentation pipeline actually creates, not every transformation that might occur in the world. If a transformation removes information, the encoder is rewarded for making the two resulting views agree despite that missing information. Information that varies unpredictably across views cannot reliably help reduce this loss. This is how

augmentation defines an implicit preference over what the representation keeps.

This can be made precise. Let $\mathcal{T}$ be a set of transformations and define the orbit of an input x as $\mathcal{O}(x) = \{T(x) : T \in \mathcal{T}\}$. If an encoder is exactly invariant, then

$$f_\theta(Tx) = f_\theta(x) \qquad \text{for every} T \in \mathcal{T}.$$

All observations in the same orbit receive the same representation. The encoder therefore acts on equivalence classes of inputs rather than on every pixel-level distinction. This is the mathematical meaning of invariance: it removes distinctions that the chosen transformation family treats as irrelevant.

That reduction is beneficial only if the target of interest is constant within each class. If Y changes when x is transformed, no exactly invariant representation can preserve all information about Y. The problem is not that optimization failed; the objective has asked the encoder to discard the distinction. In real systems the equality is approximate, but the design pressure remains.

Invariance is also different from **equivariance**. An invariant representation remains unchanged under a transformation. An equivariant representation changes in a predictable way. For example, if an image is translated, an image-level category feature may ideally remain fixed, while a spatial feature map may shift in correspondence with the image. Classification and localization therefore need not prefer the same transformation behavior. A feature map that is deliberately equivariant can be more useful than a globally invariant vector when the target depends on where a feature occurs.

Historically, visual self-supervision used hand-designed pretext tasks such as predicting image rotation (Gidaris et al., 2018), restoring the spatial order of jigsaw patches (Noroozi and Favaro, 2016), predicting spatial context (Doersch et al., 2015), and colorizing grayscale images (Zhang et al., 2016). These tasks assign synthetic targets to unlabeled data and make the target-design assumption easy to see. They remain useful conceptual examples, but the architecture survey below focuses on the broader contrastive, bootstrap, statistical, clustering and masked-prediction families that followed.

These are pretext tasks: targets are generated from the input rather than supplied by human annotation. They can teach useful features, but the target may admit shortcuts or require only a subset of the information needed downstream. Predicting image rotation, for example, need not preserve every scene distinction relevant to future prediction. Later objectives use relations between views, feature-batch statistics, or hidden content instead of one fixed synthetic label.

The principle behind representation learning

Every self-supervised method constructs a target relationship from the data. The relationship may be agreement between views, a ranking among candidates, a teacher’s output

distribution, statistics of a feature batch, a cluster assignment or a missing patch. Ask what information makes that relationship easy to satisfy, then ask whether that information is what the downstream model needs.

The map below groups methods by their dominant training signal while allowing overlap between families.

The family map in figure 3.1 groups objectives by the signal they construct, rather than by publication date.

Self-supervised objectives grouped by their training signal

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The representation contract

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The map is not a partition into disjoint categories. DINO uses teacher-student self-distillation over prototype outputs; iBOT joins masking and teacher targets; wav2vec 2.0 combines masking, quantization and contrastive discrimination.

Augmentation defines the invariances rewarded by training

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3.2. Contrastive learning

Constructing positives and negatives

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3.2.1. From pairs to candidate classification

A contrastive method begins with an anchor, a related positive example and distractors called

negatives. The encoder maps each example to a feature vector. A similarity function scores the anchor against each candidate. Training rewards the positive for receiving a higher score than the negatives.

In the common two-view construction, each source example x_n is transformed independently twice:

$$\tilde{x}_n^{(1)} = T_n^{(1)}(x_n), \qquad \tilde{x}_n^{(2)} = T_n^{(2)}(x_n).$$

The two views of the same source form a positive pair. Views from other source examples form the negative candidates for that anchor. No class label is required; source identity supplies the pair label. The label therefore means “same source under two transformations,” not necessarily “same semantic class” or “same underlying state.”

If two different source examples are semantically equivalent, treating one as another’s negative creates a false negative relative to a semantic objective. This is not an error in cross-entropy. It is a mismatch between the labels implied by pair construction and the relation we ultimately want the representation to encode.

The negative set is part of the statistical problem, not just a source of extra examples. A contrastive loss trains the encoder to preserve distinctions that help identify the positive against the sampled candidates. If negatives are easy, the model can often solve the task using broad visual cues. As training improves, the most probable negatives dominate the gradient and the model is pushed to distinguish harder cases. This can sharpen useful features, but it can also make the learned geometry depend on details of the candidate distribution.

The loss does not tell the model which differences are semantically important. It only supplies a ranking label for the candidates. If the dataset repeatedly pairs two views of the same source image while treating every other image as negative, the representation is encouraged to keep same-source views close and other sampled views relatively separated. Two different images of the same object may still be pushed apart. Methods that use multiple positives, debiased negatives or semantic labels change this constraint, but they also change the learning problem.

For two feature vectors, cosine similarity is their dot product after unit normalization:

$$s(z_i, z_j) = \frac{z_i^\top z_j}{\| z_i \|_2 \, \| z_j \|_2}.$$

A temperature $\tau>0$ scales every similarity before softmax. If two candidates have similar scores, a lower temperature magnifies their difference in the logits and makes the probability distribution more concentrated. It does not simply make every gradient larger: when the positive already has probability close to one, its cross-entropy gradient is close to zero. Temperature changes both the probability assigned to each candidate and the factor relating a logit gradient to the underlying

similarity gradient.

Suppose a batch contains N source images. Two stochastic views of each image give $2N$ encoded vectors. For anchor i, let $p(i)$ be the index of its other view. Excluding the anchor itself, every remaining view is a candidate. The NT-Xent loss for that anchor is

$$\ell_i = -\log \frac{\exp(s(z_i, z_{p(i)}) / \tau)}{\sum_{a=1}^{2N} \mathbf{1}[a \neq i] \exp(s(z_i, z_a) / \tau)}.$$

The batch loss averages this quantity over all $2N$ anchors:

[$$L_{\mathrm{NT\text{-}Xent}}=\frac{1}{2N}\sum_{i=1}^{2N}\ell_i.$$] | *stem-a16c682eaf9b33914e71c1e010c03848.svg*

In the denominator of each term are one positive and $2N-2$ negatives. The positive is included in the denominator; the anchor is not. Stating the candidate set explicitly avoids ambiguity because papers do not all use the same symbol for the number of candidates or negatives.

This objective is ordinary softmax cross-entropy. Write $a_{ij} = s(z_i, z_j) / \tau$, and let q_{ij} be the softmax probability assigned to candidate j. Differentiating with respect to each candidate logit gives

$$\frac{\partial \ell_i}{\partial a_{ij}} = q_{ij} - \mathbf{1}[j = p(i)].$$

The positive logit receives gradient $q_{i,\,p(i)} - 1$, which pushes it upward under gradient descent. A negative logit receives gradient q_{ij}, which pushes it downward. The gradient on a negative is weighted by its current probability: a hard negative receives more repulsive gradient than an already dissimilar candidate. Since $a_{ij} = s(z_i, z_j) / \tau$, differentiation with respect to the unscaled similarity introduces the factor $1 / \tau$.

For example, suppose one anchor has cosine similarities 0.80, 0.10 and -0.30 to its positive and two negatives, with $\tau = 0.5$. The logits are 1.6, 0.2 and -0.6. Softmax assigns the positive probability about 0.737, so the loss for this anchor is $-\log(0.737) \approx 0.306$. If all three candidates received equal probability, the loss would be $-\log(1/3) \approx 1.099$. The loss measures whether the correct candidate is easy to identify relative to the candidates that happened to be sampled.

For a cosine score $s(u, v) = \hat{u}^\top \hat{v}$, where $\hat{u} = u / \|u\|_2$ and $\hat{v} = v / \|v\|_2$, the derivative with respect to the first vector is

$$\frac{\partial s(u, v)}{\partial u} = \frac{1}{\|u\|_2} \left(\hat{v} - s(u, v) \hat{u} \right).$$

This gradient changes the direction of a normalized vector to improve its relative score. In a neural network, the same representation can appear as an anchor and as a candidate in other loss terms, so

its total gradient accumulates contributions from all those roles. Contrastive learning is not merely "pull positives together": it adjusts the representation so that positives outrank the negatives in the sampled candidate set.

The InfoNCE schematic visualizes the anchor, candidate set, softmax distribution and resulting gradient.

InfoNCE as candidate classification

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How the InfoNCE gradient acts on candidate scores

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3.2.2. What InfoNCE does and does not prove

InfoNCE is a relative objective. The anchor must assign a higher score to its designated positive than to the sampled alternatives. Maximizing the positive similarity alone would not impose this ranking constraint. With an unnormalized dot product the model can also increase scores by increasing vector norms; cosine normalization bounds the score but does not remove dependence on the candidate-sampling protocol.

The positive label is supplied by data construction. In a two-view image objective, it means that two views came from the same source image. A negative can still depict the same semantic class as the anchor or show a nearly identical state. This is a false negative relative to a semantic objective, not an error in the cross-entropy arithmetic. The transformation distribution creates a related risk: if a view removes a relevant feature, consistency across those views can reward the encoder for ignoring it.

InfoNCE also has an information-theoretic interpretation, but only under a specified sampling construction. Suppose one index i is selected uniformly from K candidates; y_i is sampled from the conditional distribution $p(y \mid x)$, while the other $K-1$ candidates are sampled independently from the marginal $p(y)$. Bayes' rule gives

$$P(i \mid x, y_1, \ldots, y_K) = \frac{r(x, y_i)}{\sum_{j=1}^{K} r(x, y_j)}, \qquad r(x, y) = \frac{p(y \mid x)}{p(y)}.$$

The optimal classifier therefore ranks candidates by the density ratio: a pair receives a high score

when it is more likely under the joint distribution than under independent sampling. A learned critic approximates this rule through its logits. The candidate-classification cross-entropy under this sampling construction yields the lower bound

$$I(X;Y) \geq \log K - \mathcal{L}_{\mathrm{NCE}},$$

where K includes the positive candidate. Equivalently, $\mathcal{L}_{\mathrm{NCE}} \geq \log K - I(X;Y)$. The finite candidate count limits this bound to $\log K$. The bound does not grow without limit merely because training continues.

The assumptions matter. Nearby frames, repeated examples or multiple instances of one category can violate independent negative sampling. If negatives come from a different distribution, or candidates are dependent, the same bound does not automatically follow. The density-ratio argument explains the role of the loss and its sampling assumptions; it does not turn an arbitrary contrastive loss into a certified measure of semantic information.

Finally, mutual information shared by two views is not the same as sufficiency for a later prediction. A feature can be shared between views yet irrelevant to the future quantity of interest. The information-theoretic bound says something about dependence under the sampling model, not whether all distinctions required by a downstream predictor have been retained.

3.2.3. Alignment and uniformity are different geometric requirements

For normalized representations, contrastive learning can be studied through two geometric diagnostics. **Alignment** asks matched views to be close. **Uniformity** asks independent examples not to crowd into a small part of the unit sphere. They explain different pressures in contrastive training, but they are not a universal exact decomposition of every self-supervised loss.

One alignment measure is

$$\mathcal{L}_{\mathrm{align}} = \mathbb{E}_{(x,\ x^{+}) \sim p_{\mathrm{pos}}} \| f(x) - f(x^{+}) \|_2^{\alpha},$$

where $\alpha>0$. A uniformity measure is

$$\mathcal{L}_{\mathrm{uniform}} = \log \mathbb{E}_{x,\ y \sim p_{\mathrm{data}}} \exp(-t \| f(x) - f(y) \|_2^2),$$

where $t>0$ and x,y are independent samples. Nearby embeddings contribute strongly to the exponential, so crowding raises the quantity. Wang and Isola's (2020) analysis uses normalized features and a specified sampling setup. These measures are a lens on contrastive learning, not permission to describe every SSL algorithm as optimizing the same two losses.

Uniformity is not semantic correctness: features can spread out while encoding background texture or another nuisance variable. Alignment can be excellent because an augmentation erased a task-critical distinction. A useful representation requires invariances to match the prediction problem.

Alignment and uniformity are distinct geometric requirements

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3.2.4. The energy-based view of contrastive scores

An energy-based model assigns a scalar compatibility score, or energy, to a configuration. Lower energy means the model regards the configuration as more compatible with the observed evidence. For an anchor x and candidate y, let $E_\theta(x, y)$ be the energy. A conditional distribution can be written

$$p_\theta(y \mid x) = \frac{\exp[-E_\theta(x, y)]}{\sum_{y'} \exp[-E_\theta(x, y')]}.$$

The denominator is a partition function: it normalizes the scores over all candidate values y'. If the candidate space is large or continuous, computing that normalizer can be difficult. An unnormalized energy function can still rank alternatives, but ranking alone is not the same as having an explicitly normalized probability model over the full space.

Contrastive learning can be read as energy-based candidate classification. Set $E_\theta(x, y) = -s_\theta(x, y)/\tau$. The softmax over the anchor's sampled candidates assigns lower energy to the positive and higher energy to alternatives. The NT-Xent gradient therefore has the same structure as ordinary cross-entropy over candidate energies: lower the positive's energy relative to the negatives, with more gradient assigned to negatives that currently receive high probability. The candidate denominator in InfoNCE is a finite, sampled normalization, not necessarily the full partition function over every possible image or latent.

This connection clarifies both the power and the limit of the method. A learned energy can represent a rich compatibility relation without immediately providing a tractable normalized distribution over all images. If the goal is classification among a fixed candidate set, ranking may be enough. If the goal is generation, likelihood evaluation or drawing samples from the full distribution, additional machinery is needed. Denoising diffusion models, discussed later, use a different training construction to learn a sequence of denoising transitions; they should not be treated as merely a larger InfoNCE denominator.

Energy-based learning is a general framework rather than a synonym for contrastive learning. Some energy-based methods compare observed configurations to generated or sampled alternatives; some train through likelihood or margin-based objectives. InfoNCE is one particular candidate-discrimination loss whose energy interpretation is especially transparent because its normalization and gradient are explicit. See LeCun et al. (2006) for the broader framework.

A contrastive score, an energy model and a diffusion process

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3.2.5. SimCLR and MoCo: two ways to construct candidates

SimCLR (Chen et al., 2020) made the augmented-view recipe explicit. Two views pass through an encoder f and a nonlinear projection head g, and the objective is applied to $z = g(f(x))$. The representation before the head, $h = f(x)$, is commonly retained for downstream tasks. The projector allows the loss to shape a space specialized for the pretraining comparison, rather than requiring every property of that space to be optimal for later evaluation. This is a design choice supported by the SimCLR experiments, not a theorem that a projection head always improves every representation.

SimCLR's results depended on augmentation composition, batch size, architecture, training duration and optimization. A large batch increases the number of available negatives under an in-batch objective, but changes the classification problem itself and may add more false negatives. Its contribution is a clear recipe for contrastive visual learning, not evidence that every contrastive method needs one specific batch size or projector.

MoCo (He et al., 2020) uses a query encoder and a key encoder, with a queue storing keys from earlier minibatches. In the original recipe, the query and key features are L2-normalized before their dot products are scored. The query feature $q = f_{\theta_q}(x_q)$ is trained to identify its matching key $k^+ = f_{\theta_k}(x_k)$ among the positive and queued negative keys $k_j \in Q$. Its candidate loss has the form

$$\mathcal{L}_{\mathrm{MoCo}} = -\log \frac{\exp(q^\top k^+ / \tau)}{\exp(q^\top k^+ / \tau) + \sum_{k_j \in Q} \exp(q^\top k_j / \tau)}.$$

The queue gives each query a larger candidate set than the current minibatch alone. The key branch receives no backpropagation gradient; its parameters follow an exponential moving average of the query encoder:

$$\theta_k \leftarrow m\theta_k + (1 - m)\theta_q.$$

After the loss is evaluated against the current queue, keys from the current minibatch are enqueued and the oldest keys are dequeued to keep the queue length fixed. The queue therefore contains features produced by earlier parameter values. They become stale as training proceeds; a slowly moving key encoder reduces abrupt drift in their representation space but cannot make historical keys identical to current ones. This EMA update makes the key parameters a smoothed history of the query parameters: for fixed m, contributions from earlier updates decay geometrically. MoCo

v2 (Chen et al., 2020) strengthened v1's augmentation, projection-head and optimization recipe while retaining the queue and momentum key encoder. The MoCo v1/v2 loss above therefore uses positive-plus-queue candidates. MoCo v3 (Chen et al., 2021) removes this persistent FIFO queue. Its reference implementation compares the online prediction for one crop against momentum-encoder targets for the other crop, gathered across workers. With Q_1, Q_2 denoting normalized online predictor outputs, K_1, K_2 the corresponding normalized momentum targets gathered across workers, and y the matching sample indices, its symmetric batch classification can be written

$$\mathcal{L}_{\mathrm{v3}} = \mathrm{CE}\left(\frac{Q_1 K_2^{\top}}{\tau}, y\right) + \mathrm{CE}\left(\frac{Q_2 K_1^{\top}}{\tau}, y\right).$$

The momentum encoder remains stop-gradient and EMA-updated; the online path has a predictor. This is not the v1/v2 queue algorithm with a different backbone. The reference code also multiplies each cross-entropy term by a temperature-dependent constant, a loss-scale convention rather than a different candidate relation.

Both methods remain sensitive to the definition of positive pairs and negatives. The queue expands the candidate set, while the momentum encoder mitigates abrupt feature drift without eliminating staleness. Neither mechanism establishes that every negative is semantically different.

SimCLR's two-view encoder and projection head

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Momentum contrast: query encoder, key encoder and queue

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3.2.6. Historical contrastive branches

SimCLR and MoCo are central modern examples, but their shared candidate-classification structure has earlier branches. These are not simply a sequence of equivalent losses.

Non-parametric instance discrimination (NPID) treats each image instance as a proxy class and uses a memory bank of features with a noise-contrastive objective. The signal is instance identity, which need not equal semantic identity.

PIRL asks representations to remain compatible across a pretext transformation while preserving an instance-discrimination signal. It changes the positive relation by bridging a transformed view

and the original instance. The transformation helps only when the induced invariance matches downstream needs.

Contrastive Multiview Coding (CMC) contrasts features from different views or modalities of the same source. The positive relation is a shared source rather than merely two augmentations of the same image tensor; usefulness depends on information shared between views.

AMDIM uses local and global representations at multiple feature scales. It contrasts a local feature with a global feature from the same image while mismatched images supply negatives. This introduces spatially indexed dependencies that global instance discrimination may not enforce.

These methods all construct a discrimination problem, but their positive relations and sampling distributions differ. Wu et al. (2018) study non-parametric instance discrimination; Misra and van der Maaten (2020) construct pretext-invariant representations; Tian et al. (2020) use contrastive multiview coding; and Bachman et al. (2019) introduce AMDIM. Any information-bound interpretation must follow the actual candidate sampler, not the generic label "contrastive."

Historical contrastive objectives change the positive relation

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3.2.7. CPC as a historical bridge

Contrastive Predictive Coding (CPC; van den Oord et al., 2018) encodes a sequence and uses a context vector from the past to score future latent representations. An InfoNCE-style objective teaches the context to identify the observed future latent among candidate latents. CPC is a useful historical bridge between contrastive discrimination and predictive representations, but it is not the centerpiece of a contemporary visual SSL survey.

A CPC future-latent retrieval score only shows that the correct future code can be identified among the sampled candidates under the chosen objective. It does not by itself establish accurate multi-step rollout, a sufficient predictive state, or that errors remain controlled when predictions are fed back into the model.

CPC: causal context predicts future latent candidates

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3.3. Non-contrastive objectives

Contrastive learning makes a designated positive score higher than sampled negatives. Non-contrastive methods remove that explicit competition, but they still need a mechanism that prevents agreement from becoming trivial. BYOL and SimSiam use asymmetric prediction pathways; DINO matches teacher distributions; Barlow Twins and VICReg constrain batch statistics.

3.3.1. Why agreement alone admits collapse

Suppose two views of the same input are trained only to produce equal representations. A simple invariance loss is

$$L_{\text{inv}} = \frac{1}{N}\sum_{i=1}^{N} \| z_i - z_i' \|_2^2 .$$

If the encoder maps every input to the same constant vector c, then $z_i = z_i' = c$ for every example and the loss is zero. The objective has no term requiring different inputs to remain distinguishable. A constant output is therefore a collapsed solution, even though it is useless for representing the data.

This observation does not mean every method that matches two views will collapse. It means agreement alone cannot rule collapse out. The architecture, stop-gradient placement, target updates, normalizations and optimization dynamics affect which solutions training reaches. The methods below differ in precisely those mechanisms; none should be described as having a universal guarantee based only on its loss formula.

It is useful to distinguish complete collapse from dimensional collapse. In complete collapse, all inputs map to the same vector and the representation contains no information about differences between examples. In dimensional collapse, the output varies, but most of its covariance is concentrated in a small number of directions. A model can therefore avoid the constant solution and still use its nominal d-dimensional space inefficiently.

These cases require different diagnostics. Per-coordinate variance detects a constant dimension; the spectrum of the covariance matrix shows whether variation is spread across many directions. But neither diagnostic answers whether those directions contain the information of interest. A feature can vary strongly with background texture and have high effective rank while carrying little information about the target. Anti-collapse is necessary for useful representations, not sufficient.

The covariance spectrum gives a more quantitative description of dimensional collapse. Let $\lambda_1, \ldots, \lambda_d$ be the nonnegative eigenvalues of the feature covariance matrix. Normalize them into proportions

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \qquad r_{\text{eff}} = \exp\left(-\sum_i p_i \log p_i\right).$$

The quantity r_{eff} is an effective-rank diagnostic. If all variance lies in one direction, the distribution p_i is concentrated and the effective rank is near one. If variance is spread evenly across d directions, the effective rank is d. This statistic differs from the count of nonzero dimensions: a representation can have a formally full-rank covariance matrix while almost all its variance lies in a few directions.

Effective rank remains a geometric diagnostic, not a semantic one. A high-rank representation can encode color noise, background texture or image identity; a low-rank representation can still preserve all information required for a simple target. Variance, covariance and effective rank tell us about the distribution of features across a batch. They do not identify what the features mean or prove that a selected future variable is recoverable. That requires a separate target-specific evaluation.

Why agreement alone permits collapse

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3.3.2. BYOL and SimSiam: bootstrap prediction

BYOL (Grill et al., 2020) has an online encoder, a projection head and a predictor, alongside a target encoder and target projection head. Only the online network receives gradients. The target network is updated by an exponential moving average (EMA) of the online encoder and projector; it is not updated by backpropagation.

Let $z_1 = g_\phi(f_\theta(x_1))$, $p_1 = q_\psi(z_1)$, and $\bar{z}_2 = g_{\bar{\phi}}(f_{\bar{\theta}}(x_2))$. The prediction loss for one direction is the normalized regression

$$D(p_1, \text{sg}(\bar{z}_2)) = 2 - 2\frac{p_1^\top \bar{z}_2}{\| p_1 \|_2 \, \| \bar{z}_2 \|_2},$$

where sg denotes stop-gradient. A second term swaps the views. The online predictor is trained to match the target feature, while the target feature is treated as fixed for that gradient calculation.

The target parameters change according to

$$\bar{\theta} \leftarrow m\bar{\theta} + (1-m)\theta, \qquad \bar{\phi} \leftarrow m\bar{\phi} + (1-m)\phi,$$

where m is the momentum coefficient, usually close to one.

For clarity, the two-view objective can be written

$$L_{\text{BYOL}} = D(p_1, \text{sg}(\bar{z}_2)) + D(p_2, \text{sg}(\bar{z}_1)).$$

During a gradient step, the first term updates the online encoder, projector and predictor through p_1 while the target value $\bar{z}_2$ is fixed. The second term swaps the views. After the gradient step, EMA moves the target toward the updated online parameters. Across repeated updates, older online weights have geometrically decreasing influence; at $m = 0$ the target copies the online network, while larger m slows its change. This temporal smoothing is part of BYOL's optimization dynamics, not a universal proof against collapse.

SimSiam (Chen and He, 2021) uses the same encoder and projector for both views and a predictor on each branch, but no EMA target network. With $z_i = g(f(x_i))$ and $p_i = q(z_i)$, its symmetric loss has the form

$$L_{\text{SimSiam}} = D_{\text{SimSiam}}(p_1, \text{sg}(z_2)) + D_{\text{SimSiam}}(p_2, \text{sg}(z_1)),$$

where D_{SimSiam} is negative cosine similarity. The stop-gradient leaves a target's forward value unchanged but removes the gradient path through that target in the corresponding term. Because the loss is symmetric, each shared branch acts as predictor in one term and target in the other. BYOL instead maintains a separate target network whose parameters follow the EMA update.

The predictor and stop-gradient change optimization dynamics, but the constant-output solution remains a solution of the similarity objective. The distance used in the symmetric objective is

$$D_{\text{SimSiam}}(p, z) = -\frac{p^\top z}{\| p \|_2 \, \| z \|_2}.$$

Minimizing this negative cosine similarity maximizes directional agreement without requiring the two vectors to have the same norm. In $D(p_1, \text{sg}(z_2))$, the numerical value of z_2 still affects the loss, but its target-side gradient is removed. Because the loss is symmetric and both views share an encoder, that view still contributes encoder gradients when it serves as the predictor-side representation in the opposite term. The original SimSiam study found stop-gradient essential to its tested recipe; this is not a theorem that every detached Siamese network avoids collapse.

BYOL and SimSiam: similar predictors, different target updates

Artwork placeholder: Figure 3.14. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F14_Dadwal.md.

3.3.3. DINO: match a teacher's distribution

DINO (Caron et al., 2021) matches distributions rather than regressing one feature vector onto

another. The student and teacher map different views to logits over learned prototype outputs. If $a_s(x)$ and $a_t(x)$ are their logits, the student and teacher distributions are

$$p_s(x) = \mathrm{softmax}\left(\frac{a_s(x)}{T_s}\right), \qquad q_t(x) = \mathrm{softmax}\left(\frac{a_t(x) - c}{T_t}\right).$$

Here c is a running center of teacher outputs, T_t controls teacher-target sharpness, and the teacher parameters follow an EMA of the student parameters. The student is trained across selected, different teacher-student views. For a target distribution q_t, the student's cross-entropy is

$$H(q_t, p_s) = -\sum_{k=1}^{K} q_{t,k} \log p_{s,k}.$$

Training sums this loss over selected teacher-student crop pairs.

Cross-entropy decomposes into the entropy of the target plus a divergence:

$$H(q_t, p_s) = H(q_t) + D_{\mathrm{KL}}(q_t \,\|\, p_s).$$

For a fixed teacher target, the entropy term is constant with respect to the student, so minimizing cross-entropy makes the student distribution approach the teacher distribution. The decomposition also clarifies a collapse risk. If the teacher produces the same distribution for every input, the student can match it with a small KL divergence while learning little input-dependent structure. A low loss by itself does not show that the teacher's targets vary meaningfully across examples.

Centering and sharpening address different undesirable target distributions. Without centering, one output coordinate can dominate across many inputs, encouraging the teacher to repeatedly favor the same prototype. Centering counteracts this dominance, but by itself may encourage overly uniform outputs. Sharpening lowers the teacher temperature, making each target distribution more concentrated rather than uniform. Their combination helps retain input-dependent assignments. These are practical anti-collapse mechanisms of the DINO recipe; they should not be generalized into a theorem that centering and sharpening alone guarantee useful features in every architecture or training setup.

Prototype indices are learned output coordinates, not automatically human-defined categories. DINO is best understood as self-distillation with a prototype distribution, rather than as an ordinary clustering algorithm.

Multi-crop training supplies a useful extra constraint. The teacher receives the larger global crops, while the student also sees smaller local crops. A local crop usually lacks some of the global context, so its output must match a target produced from a different field of view. This encourages features to remain consistent across scales, but a local crop may not contain enough evidence to infer every property of the full scene. The crop protocol determines which consistency the

objective rewards.

These releases also illustrate a general caution: reported improvements belong to a complete training system, including data, architecture, optimization and evaluation, rather than to an isolated loss substitution. The following details separate the original DINO mechanism from components added in later recipes.

DINOv2 is a combined training recipe, not simply a replacement loss: it combines class-token DINO distillation, iBOT-style masked patch distillation and a KoLeo nearest-neighbour regularizer. The class branch matches teacher distributions across crops; the patch branch predicts teacher prototype distributions at masked student positions; KoLeo discourages local crowding in normalized backbone features. Depending on configuration, targets may use centering or Sinkhorn-Knopp balancing. Centering subtracts a running mean of teacher logits; Sinkhorn-Knopp iteratively balances assignment marginals. They are not synonyms.

DINOv3 retains related distillation machinery and adds large-scale data, architecture and training choices. Gram anchoring, when enabled, matches teacher and student pairwise patch-similarity matrices rather than requiring coordinate-by-coordinate equality of feature vectors. It is a relational constraint, not proof of physical-state sufficiency, and it is configuration-dependent. The DINOv2 and DINOv3 papers describe broader training recipes; their results should not be attributed to Gram anchoring alone without controlled ablations.

DINO teacher-student cross-view distillation

Artwork placeholder: Figure 3.15. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F15_Dadwal.md.

3.3.4. Redundancy reduction and explicit variance constraints

BYOL, SimSiam and DINO make training dynamics or teacher targets central to avoiding trivial solutions. Barlow Twins (Zbontar et al., 2021) and VICReg (Bardes et al., 2022) express another idea: require paired views to agree, but constrain the statistics of features across a batch.

3.3.5. Barlow Twins: cross-correlation toward identity

Let Z^A and Z^B be feature batches from two views. For dimension i, let μ_i^A and σ_i^A be its mean and standard deviation across the batch, and likewise for view B. Standardize each dimension:

$$\tilde{z}^A_{b,\,i} = \frac{z^A_{b,\,i} - \mu_i^A}{\sigma_i^A + \epsilon}, \qquad \tilde{z}^B_{b,\,i} = \frac{z^B_{b,\,i} - \mu_i^B}{\sigma_i^B + \epsilon}.$$

Then define the cross-correlation matrix C:

$$C_{ij} = \frac{1}{N} \sum_{b=1}^{N} \tilde{z}^{A}_{b,\,i} \tilde{z}^{B}_{b,\,j}.$$

If corresponding dimensions agree and distinct dimensions are uncorrelated across views, C approaches the identity matrix. Barlow Twins minimizes the diagonal mismatch and penalizes off-diagonal cross-correlation:

$$L_{\mathrm{BT}} = \sum_{i} (1 - C_{ii})^2 + \lambda \sum_{i} \sum_{j \neq i} C_{ij}^2.$$

The diagonal terms encourage each coordinate to agree with its counterpart across views; the off-diagonal terms penalize linear cross-view correlation between different coordinates. Because features are standardized over the batch, these entries describe co-variation across examples rather than the relationship between two coordinates in one example. Duplicate coordinates incur an off-diagonal penalty, while poorly aligned coordinates incur a diagonal penalty.

The coefficient λ sets the strength of decorrelation relative to alignment. This is a second-order constraint: uncorrelated features can still have nonlinear dependence, and the coordinates are not automatically meaningful physical variables. Barlow Twins specifies a pattern for the batch cross-correlation matrix, not merely a distance between paired vectors.

Barlow Twins: the cross-correlation matrix

Artwork placeholder: Figure 3.16. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F16_Dadwal.md.

3.3.6. VICReg: make three constraints explicit

VICReg separates invariance, variance and covariance. For two feature batches Z and Z', the invariance term is a mean-squared difference:

$$s(Z, Z') = \frac{1}{N} \| Z - Z' \|_F^2 .$$

For each feature dimension j, let $\sigma_j(Z)$ be its standard deviation over the batch. A variance penalty puts a floor under each dimension's standard deviation:

$$v(Z) = \frac{1}{d} \sum_{j=1}^{d} \max\left(0, \gamma - \sqrt{\mathrm{Var}(Z_j) + \epsilon}\right).$$

The covariance term penalizes the squared off-diagonal entries of the empirical covariance matrix:

$$c(Z) = \frac{1}{d} \sum_{i \neq j} \mathrm{Cov}(Z)_{ij}^2.$$

The complete objective combines the three terms:

$$L_{\mathrm{VICReg}} = \lambda s(Z, Z') + \mu [v(Z) + v(Z')] + \nu [c(Z) + c(Z')],$$

where λ, μ, ν set their relative weights. The invariance term aligns paired views. The variance term penalizes a feature dimension whose standard deviation falls below γ. The covariance term discourages linear redundancy between different dimensions. The original VICReg variance term uses an unsquared hinge. Once a feature's standard deviation reaches the margin, that dimension contributes zero variance penalty; the objective enforces a floor rather than rewarding ever-increasing variance.

The sample covariance used for $c(Z)$ is computed after centering the feature columns. For a batch $Z \in \mathbb{R}^{N \times d}$, let $\mu_j = N^{-1} \sum_{b=1}^{N} z_{b,\,j}$. Its covariance entries are

$$\mathrm{Cov}(Z)_{ij} = \frac{1}{N-1} \sum_{b=1}^{N} (z_{b,\,i} - \mu_i)(z_{b,\,j} - \mu_j).$$

The variance penalty and covariance penalty are deliberately different. A coordinate can have sufficient variance while duplicating another coordinate; covariance regularization penalizes that redundancy. A coordinate can have low covariance with every other coordinate yet be nearly constant; the variance penalty addresses that case. The invariance term then asks the two views to agree. These terms are complementary, not interchangeable.

The reductions are part of the definition. In $s(Z, Z')$, the squared feature-vector difference is summed over coordinates and averaged over examples; it is not automatically the same scalar as elementwise mean-squared error that divides by both batch size and feature dimension. That difference rescales the invariance term by the embedding dimension, changing the effective balance between λ, μ and ν. When comparing equations and implementations, check whether the norm is summed per example or averaged per coordinate before interpreting the coefficients.

A fully constant representation makes $s(Z, Z') = 0$ and makes all off-diagonal covariances zero, but it also makes each standard deviation zero. For any positive margin γ, the variance term is therefore positive. In this sense, the variance floor makes exact constant collapse incompatible with minimizing all three terms simultaneously. This explicit penalty differs from BYOL or SimSiam, where a constant representation can make the view-agreement part of the objective small. It still does not guarantee that optimization never reaches a bad solution or that nonconstant directions encode the desired semantics.

The methods are related but not identical. Barlow Twins uses normalized cross-correlation as the

central matrix objective. VICReg uses a direct invariance term, a standard-deviation floor and a covariance penalty. Their shared lesson is that view agreement and non-collapse are separate requirements. Their moment constraints are not a guarantee of a world-model-ready state.

A simple two-coordinate example makes the distinction concrete. If both coordinates are copies of the same varying scalar, their variances may exceed the margin, so the variance term is satisfied; however, their covariance is high and the covariance term penalizes them. If the second coordinate is constant, its covariance with the first may be zero, but its standard deviation is below the margin and the variance term penalizes it. Only a feature batch with adequate spread and sufficiently low linear redundancy can make both statistical penalties small. The remaining variation can still describe the wrong thing, so a separate task-based evaluation is necessary.

Listing 3.1 Three VICReg terms in the official PyTorch implementation (source excerpt)

```
# Source: facebookresearch/vicreg/main_vicreg.py
repr_loss = F.mse_loss(x, y)

x = x - x.mean(dim=0)
y = y - y.mean(dim=0)
std_x = torch.sqrt(x.var(dim=0) + 0.0001)
std_y = torch.sqrt(y.var(dim=0) + 0.0001)
std_loss = (
    torch.mean(F.relu(1 - std_x)) / 2
    + torch.mean(F.relu(1 - std_y)) / 2
)

cov_x = (x.T @ x) / (self.args.batch_size - 1)
cov_y = (y.T @ y) / (self.args.batch_size - 1)
cov_loss = off_diagonal(cov_x).pow_(2).sum().div(self.num_features) + \
           off_diagonal(cov_y).pow_(2).sum().div(self.num_features)

loss = (self.args.sim_coeff * repr_loss
        + self.args.std_coeff * std_loss
        + self.args.cov_coeff * cov_loss)
```

Long expressions are wrapped to fit the book’s listing width without changing the calculation. The official PyTorch implementation uses view agreement, a standard-deviation hinge with target one, and squared off-diagonal covariance. The helper and surrounding architecture are in the project’s main training script. The upstream repository uses the MIT License; exact reductions and weights should be matched when reproducing a result.

VICReg separates agreement, variance and covariance

Artwork placeholder: Figure 3.17. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F17_Dadwal.md.

Core non-contrastive mechanisms at a glance

Artwork placeholder: Figure 3.18. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F18_Dadwal.md.

3.3.7. Beyond covariance: SIGReg, LeJEPA and VISReg

VICReg constrains variance and off-diagonal covariance. These are interpretable second-order statistics, but two distributions can share a mean and covariance while differing in multimodality or tail behavior. SIGReg and VISReg investigate distribution-shape constraints, particularly for joint-embedding predictive training.

LeJEPA's theory argues that an isotropic Gaussian embedding distribution is optimal for downstream prediction risk under the assumptions developed in its paper. For a standard Gaussian $Z \sim \mathcal{N}(0, I_d)$, every unit-direction projection $w^\top Z$ is standard normal. This is a theory result under specified conditions, not a universal claim that every downstream task requires Gaussian latents.

SIGReg approximates distribution matching with random one-dimensional projections. For projected samples $u_1, \ldots, u_B$, the empirical characteristic function is

$$\widehat{\varphi}_B(t) = \frac{1}{B}\sum_{b=1}^{B} e^{itu_b}, \qquad \varphi_{\mathcal{N}(0,\,1)}(t) = e^{-t^2/2}.$$

For random unit directions $w_1, \ldots, w_K$, a schematic form of this discrepancy is

$$\mathcal{L}_{\mathrm{SIGReg}}(Z) = \frac{1}{K}\sum_{k=1}^{K} \int_{\mathbb{R}} \left| \frac{1}{B}\sum_{b=1}^{B} e^{itw_k^\top z_b} - e^{-t^2/2} \right|^2 \omega(t)\, dt,$$

where $\omega(t)$ is the weighting used by the prescribed Epps-Pulley-style test. The practical implementation approximates the integral with its specified quadrature points and weights. The Cramér-Wold theorem motivates the sketching argument: all one-dimensional projections together characterize a distribution. A finite collection of projections is only an approximation, not proof of exact multivariate Gaussianity. Quadrature, weighting and normalization must follow the source.

LeJEPA combines a predictive objective with SIGReg. It constructs a target from the mean embeddings of global views of the same source instance, then predicts that target from other views. Views must remain grouped by original image; averaging across images defines a different objective. A schematic objective is

$$\mathcal{L}_{\mathrm{LeJEPA}} = (1-\lambda)\mathcal{L}_{\mathrm{pred}} + \lambda\mathcal{L}_{\mathrm{SIGReg}}.$$

The paper's recipe is designed not to require an EMA target encoder and stop-gradient target

branch in the same way as BYOL/SimSiam. That is a different design, not evidence that earlier asymmetries are unnecessary for every task.

VISReg separates scale, shape and center. Keep two distinct meanings of **center** separate: the prediction target is the mean embedding of global views of the same source image; the regularizer's center term penalizes the minibatch mean for each view.

In the reference implementation inspected for this chapter, the projected embeddings have shape $Z \in \mathbb{R}^{V \times B \times d}$, with V views, minibatch size B and feature dimension d. For each view v, the batch mean and centered embeddings are

$$\mu_{v,\,j} = \frac{1}{B}\sum_{b=1}^{B} z_{v,\,b,\,j}, \qquad \hat{z}_{v,\,b,\,j} = z_{v,\,b,\,j} - \mu_{v,\,j}.$$

The implementation's coordinate scale is the population standard deviation with a small numerical offset:

$$\sigma_{v,\,j} = \sqrt{\frac{1}{B}\sum_{b=1}^{B} \hat{z}_{v,\,b,\,j}^{\,2}} + \epsilon.$$

Its scale term pulls those standard deviations toward one:

$$\mathcal{L}_{\text{scale}} = \frac{1}{Vd}\sum_{v=1}^{V}\sum_{j=1}^{d} (\sigma_{v,\,j} - 1)^2.$$

For the shape term, the centered features are normalized by a **detached** copy of the scale vector. The stop-gradient is applied to the denominator, and no second epsilon is added after that scale has already been stabilized:

$$\tilde{z}_{v,\,b,\,j} = \frac{\hat{z}_{v,\,b,\,j}}{\operatorname{sg}(\sigma_{v,\,j})}.$$

For each randomly sampled unit direction w_k, project the normalized features and sort the B values. Let $q_b = \Phi^{-1}(b/(B+1))$ be the corresponding standard-normal quantiles for $b = 1, \ldots, B$. The reference implementation averages the squared differences over views, projection directions and batch positions:

$$\mathcal{L}_{\text{shape}} = \frac{1}{VKB}\sum_{v=1}^{V}\sum_{k=1}^{K} \left\| \operatorname{sort}(\tilde{Z}_v w_k) - q_{1:B} \right\|_2^2.$$

The batch-center penalty is

$$\mathcal{L}_{\text{center}} = \frac{1}{Vd} \sum_{v=1}^{V} \| \mu_v \|_2^2 .$$

The complete regularizer keeps three explicit weights:

$$\mathcal{L}_{\text{Reg}} = \lambda_{\text{scale}} \mathcal{L}_{\text{scale}} + \lambda_{\text{shape}} \mathcal{L}_{\text{shape}} + \lambda_{\text{center}} \mathcal{L}_{\text{center}}.$$

The full reference recipe also has a separate **view-invariance** term. For each source image b, let c_b be the mean of its global-view projections. The model matches every global and local projection $z_{v,\,b}$ to that same-image center:

$$\mathcal{L}_{\text{inv}} = \frac{1}{VBd} \sum_{v=1}^{V} \sum_{b=1}^{B} \| z_{v,\,b} - c_b \|_2^2 , \qquad c_b = \frac{1}{V_g} \sum_{v \in \mathcal{V}_g} z_{v,\,b}.$$

The training loss mixes this invariance term with the regularizer, $\mathcal{L} = (1 - \lambda_{\text{mix}}) \mathcal{L}_{\text{inv}} + \lambda_{\text{mix}} \mathcal{L}_{\text{Reg}}$.

The **same-image global-view center** is the target for view consistency, whereas the **per-view minibatch mean** is penalized by $\mathcal{L}_{\text{center}}$. They are different operations. SIGReg instead uses a characteristic-function discrepancy; VISReg compares sorted projected samples with Gaussian quantiles. The stop-gradient here decouples scale and shape gradients; it is not a BYOL-style teacher pathway.

The VISReg authors report stronger correction near collapse and favorable results on tested out-of-distribution, low-rank and long-tailed settings. These are reported findings, not guarantees for every optimizer or data distribution. Report batch size, projection count, objective weights, architecture and evaluation protocol when comparing these methods. Balestriero and LeCun's LeJEPA paper and the VISReg paper describe the respective recipes and assumptions.

VICReg, SIGReg/LeJEPA and VISReg constrain different aspects of feature distributions

Artwork placeholder: Figure 3.19. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F19_Dadwal.md.

3.3.8. Other methods in brief

DeepCluster alternates between clustering the current feature vectors and training the encoder to predict the resulting cluster assignments. The pseudo-labels therefore change as the representation changes. **SwAV** computes prototype assignments online and predicts one view's assignment from another view's features; a balancing procedure discourages all examples from being assigned to a small number of prototypes. Both methods turn unlabeled features into pseudo-targets, but differ in how assignments are generated and refreshed.

The table is an index, not a claim that these systems share one objective. In image-text training, **CLIP** (Radford et al., 2021) uses normalized image and text embeddings, a learned logit scale and cross-entropy in both image-to-text and text-to-image directions. **SigLIP** (Zhai et al., 2023) uses pairwise sigmoid/logistic terms for matched and mismatched pairs rather than the same batchwise softmax. Natural-language descriptions are supervision from data even though they are not manually assigned class labels.

For speech, **wav2vec 2.0** (Baevski et al., 2020) masks latent speech frames and contrasts context features against quantized targets; **HuBERT** (Hsu et al., 2021) predicts cluster-derived targets at masked positions; **data2vec** (Baevski et al., 2022) predicts contextualized teacher features. For language, **BERT** (Devlin et al., 2019) predicts selected corrupted tokens from bidirectional context, GPT-style causal language models (Radford et al., 2019) predict the next token from a prefix, and **ELECTRA** (Clark et al., 2020) discriminates observed tokens from generator-replaced tokens. Joint-embedding predictive architectures (JEPAs), including I-JEPA and V-JEPA, predict learned target representations from context rather than reconstructing pixels; their architectures and variants are covered in Chapter 4. These examples differ in both the context available to the model and the target it must predict. Diffusion and energy-based models have different generative or probabilistic contracts and should not be folded into a single family merely because their training signals are data-derived.

Table 3.1 Self-supervised objectives compared by the signal they construct

Method family	**What the objective enforces**	**What it does not guarantee**
SimCLR / MoCo	Identify a positive among sampled candidates	Negatives may be semantically related; features depend on candidate construction
CPC	Rank a future latent against sampled candidates	Candidate identification is not a calibrated transition model or an action-conditioned rollout
BYOL / SimSiam	Predict a detached view representation	Agreement alone does not rule out constant outputs
DINO family	Match teacher distributions over learned prototypes	Teacher targets can collapse or omit information of interest
Barlow Twins	Cross-view correlation matrix near identity	Second-order decorrelation is not semantic sufficiency
VICReg	View agreement plus variance and covariance constraints	Nonconstant, decorrelated features can still encode nuisance variation
SIGReg / VISReg	Constrain projected feature-distribution shape	Distributional regularity does not establish semantic or predictive sufficiency
MAE	Predict pixel values at masked patches	Pixel reconstruction does not prove predictive sufficiency
BEiT / iBOT / data2vec	Predict discrete codes or teacher features	The representation inherits the chosen target’s biases
JEPAs	Predict target embeddings from context	Feature prediction does not automatically yield a sufficient state
DeepCluster / SwAV	Predict or match pseudo-cluster assignments	Cluster assignments need not correspond to downstream concepts

DeepCluster and SwAV: pseudo-labels and prototypes

Artwork placeholder: Figure 3.20. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F20_Dadwal.md.

Different modalities create different self-supervised targets

Artwork placeholder: Figure 3.21. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F21_Dadwal.md.

3.4. Masked prediction: choose what the model must recover

3.4.1. One mask, several target spaces

Suppose an input is divided into visible positions V and masked positions M. An encoder processes the allowed context and a prediction head estimates a target y_i at masked positions:

$$L = \frac{1}{|M|} \sum_{i \in M} \ell(\hat{y}_i,\ y_i).$$

This form leaves the crucial choice open: what is the target? It might be pixel values, a discrete code, a word token, a replaced-token label or a contextualized feature. The mask tells the model where information is withheld; the target tells it what relationship must be learned.

For a probabilistic predictor, the general objective can be written as a conditional negative log-likelihood:

$$L(\theta) = -\mathbb{E}_{X_V,\ Y_M}[\log p_\theta(Y_M \mid X_V)],$$

where X_V is the visible context and Y_M is the hidden target. If Y_M is a discrete token or code, this becomes cross-entropy. If the model uses a Gaussian likelihood with fixed variance for a continuous target, minimizing negative log-likelihood is equivalent, up to constants and scale, to minimizing squared error. Under squared error, the optimal predictor in expectation is the conditional mean $\mathbb{E}(Y_M \mid X_V)$. Under cross-entropy, the optimal predicted class distribution is the conditional distribution of the target given the visible context.

The distinction appears directly in the loss. With squared-error regression, a predictor is rewarded for numerical proximity to a target vector. With cross-entropy, it is rewarded for assigning probability to the correct discrete label or teacher distribution. These losses define different geometries of error. For a discrete token, predicting a related code is still wrong unless the correct code receives probability; for a continuous target, errors are penalized according to their

magnitude in the chosen feature space.

Masking also changes what the model is allowed to observe. A model trained to reconstruct a hidden patch from nearby visible pixels can exploit local texture continuity. A model trained to predict a contextualized teacher feature must reproduce a target shaped by the teacher's wider context and representation space. Neither target is neutral: it specifies which aspects of the missing content distinguish a good prediction from a bad one.

The target-space schematic holds the mask fixed and changes only the prediction target.

One visible context, several masked-prediction targets

Artwork placeholder: Figure 3.22. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F23_Dadwal.md.

3.4.2. MAE: predict masked pixels

A masked autoencoder (MAE; He et al., 2022) divides an image into patches and hides a large fraction of them. Only visible patches pass through the encoder. A smaller decoder receives the encoded visible patches and learned mask tokens inserted at the missing positions; it predicts pixel values at the original patch locations. The asymmetric design saves encoder computation because the encoder never processes the hidden tokens.

Let $m_i = 1$ when patch i is masked and 0 otherwise, with patch target x_i and prediction $\hat{x}_i$. The reconstruction objective is

$$L_{\text{MAE}} = \frac{\sum_i m_i \, \| \, x_i - \hat{x}_i \, \|_2^2}{\sum_i m_i}.$$

The normalization is over masked patches, so visible patches do not contribute to the loss. Under squared error, the conditional mean is the best prediction in expected squared-error terms. If several patch contents are plausible from the visible context, the conditional mean can blur those alternatives. Pixel reconstruction can nevertheless learn useful features; its limitation is that pixel accuracy does not guarantee that the retained encoder feature preserves every variable needed for a later prediction. The encoder, rather than the reconstruction decoder, is usually the component transferred to a downstream task.

The original MAE recipe uses a high masking ratio, 75% of image patches. The encoder therefore processes a sparse visible subset, reducing token-processing cost. The mask ratio also controls prediction difficulty: with few masked patches, more of the image is directly available as context; with too many, the visible content may not determine the missing patches well. The 75% setting is characteristic of the original recipe, not a universal optimum for every resolution, patch size,

decoder or target.

The architecture in 3.23 shows why the MAE encoder sees only visible patches while its decoder reconstructs masked positions.

MAE's visible-only encoder and masked-patch decoder

Artwork placeholder: Figure 3.23. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F24_Dadwal.md.

3.4.3. BEiT, iBOT and data2vec: change the target

BEiT (Bao et al., 2022) predicts discrete visual-token IDs produced by a learned tokenizer. Rather than regress pixel values, the model classifies the tokenizer's code for each masked patch. The tokenizer introduces its own prior: useful structure can be learned from the code vocabulary, but tokenizer errors and biases enter the target.

iBOT (Zhou et al., 2022) combines masked patch prediction with teacher-student self-distillation. SimMIM (Xie et al., 2022) uses pixel reconstruction; MaskFeat (Wei et al., 2022) predicts visual features such as HOG; BEiT v2 (Peng et al., 2022) improves the discrete-token route with a semantic visual tokenizer. These methods change the target while keeping the general idea of withholding input content. The target families can be written schematically as

$$L_{\text{BEiT}} = -\log p_\theta(k_i \mid x_{\text{visible}}), \qquad L_{\text{iBOT}} = -\sum_k q_{t,\,i,\,k} \log p_{s,\,i,\,k},$$

where k_i is the tokenizer's discrete code for a masked patch and $q_{t,\,i}$ is a teacher distribution over prototype outputs for patch i. BEiT assigns probability to a discrete target; iBOT matches a teacher distribution. For data2vec, a contextualized teacher vector y_i^t supplies a continuous target and the student predicts that vector under a regression loss such as

$$L_{\text{data2vec},\,i} = \| \hat{y}_i - y_i^t \|_2^2 .$$

These schematic expressions show the target-space distinction; each paper has further implementation choices, including the teacher, normalization, masking protocol and the subset of positions used in its loss. They are not full reproductions of those objectives. The differences reinforce the central point: the mask alone does not define the learning task.

These objectives predict a target **conditioned on what remains visible**. They do not, by themselves, specify a full generative model of the input distribution. A masked predictor may estimate a conditional mean, a discrete target distribution or a teacher feature. It need not define a normalized joint distribution over every possible image or support sampling a complete image. A generative model has additional requirements: its factorization or sampling process must define

how a complete observation can be generated. This distinction prevents “reconstructs a patch” from being confused with “learns a generative model of images.”

The target also controls the granularity of the error signal. Pixel regression distinguishes exact pixel values and penalizes large numerical deviations more than small ones. Discrete-code cross-entropy cares whether probability is assigned to the correct code, with similarities between codes not automatically reflected in the loss. Teacher-feature regression uses distances in a learned embedding space, so two different target patches may be considered close if the teacher represents them similarly. No target is universally best: the choice decides what counts as a meaningful prediction error.

data2vec (Baevski et al., 2022) predicts contextualized teacher features from masked input and was designed as a framework spanning speech, vision and language. A teacher feature has already aggregated context and training biases from the teacher; it is not simply the original pixel content transformed by a fixed operation. This can make the target more structured, but the student also inherits omissions and biases in the teacher’s feature space.

BERT-style masked language modeling predicts selected corrupted tokens from bidirectional context. GPT-style causal language modeling predicts the next token from preceding tokens only. ELECTRA trains a discriminator to detect replaced tokens. These are brief analogies for target/context construction, not additional implementation tutorials.

Discrete tokens and teacher features are different masked targets

Artwork placeholder: Figure 3.24. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F25_Dadwal.md.

3.4.4. Masked prediction and diffusion solve different problems

Masked prediction and diffusion both infer targets from corrupted inputs, but they define different tasks. A masked objective selects positions and predicts their hidden contents from the visible context. In MAE the target is pixels; BEiT uses discrete codes; iBOT and data2vec use teacher-derived targets.

Diffusion instead corrupts a sample at a chosen noise level and trains a denoiser to estimate the corruption or an equivalent score-related target. A schedule of noise levels and a reverse denoising process turn that objective into a generative sampling procedure. A masked-prediction encoder is commonly transferred after its decoder is removed; predicting missing patches does not, by itself, define a generative model over complete images. Conversely, a diffusion model may learn useful internal features, but its denoising objective does not guarantee a compact state sufficient for a downstream prediction.

The key distinction here is between the target used to train a representation and the process required to generate complete samples. Diffusion parameterizations, noise schedules and sampling belong to the book's later treatment of generative models.

Masked prediction and diffusion learn different conditional tasks

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3.5. Latent bottlenecks

The projection head, global vector and local feature grid

Artwork placeholder: Figure 3.26. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F27_Dadwal.md.

3.5.1. Dimension is capacity, not useful information

A latent bottleneck maps a high-dimensional observation or token grid into a constrained representation. The simplest form is a vector $h \in \mathbb{R}^d$. Lowering d limits the number of coordinates, but dimension alone does not measure useful information. A small vector may preserve the few variables needed for a prediction; a wider vector may devote many dimensions to texture, illumination or other nuisance variation.

The information-bottleneck principle expresses this trade-off using mutual information. Let X be the observation, Z its representation and Y a target relevant to the later task. A conceptual objective is

$$\min I(X; Z) - \beta I(Z; Y).$$

The first term penalizes information about the input retained by Z; the second rewards information about Y. The coefficient β controls the trade-off. This is a theoretical lens, not the loss explicitly optimized by every method in this chapter. Mutual information for deterministic continuous representations requires care; practical SSL objectives usually optimize a tractable proxy.

The information bottleneck gives a useful idealization of **minimal sufficiency**. If Z is a deterministic function of X and is sufficient for Y, then $Y \perp X \mid Z$, so

$$I(Z; Y) = I(X; Y).$$

The representation has retained all information that X provides about Y, even if it has discarded many details of X unrelated to Y. A minimal sufficient representation seeks to retain that predictive

information while minimizing what it carries about the rest of the input. The information-bottleneck objective expresses a trade-off, not a direct recipe for finding a unique vector. Different representations can be equally sufficient, especially when features are transformed by invertible maps.

The important question is not “how small can the vector be?” It is “which distinctions can be discarded without changing the predictions we need to make?” A representation useful for image retrieval may be insufficient for predicting motion if it merges observations with different velocities or hidden states.

This is why raw dimension is not the same as information rate. A real-valued vector with a small number of coordinates can, in principle, encode many distinctions in very fine numerical differences. In practice, finite precision, noise, limited data and the smoothness of learned functions constrain how reliably such distinctions can be used. Conversely, a very wide feature may carry highly redundant coordinates. Dimension is a proxy for capacity, while useful compression concerns which differences remain reliably decodable under the conditions where the model will be used.

3.5.2. The projection head and the feature that survives

Many contrastive and bootstrap methods use a projection head between an encoder and the objective:

$$h = f_\theta(x), \qquad z = g_\phi(h).$$

The loss is calculated on z, while the representation evaluated downstream is often h. A projection head gives the pretraining objective a dedicated space in which to impose its constraints. It does not follow that z and h have the same geometry or serve the same downstream purpose.

The exported tensor must be identified explicitly. Some evaluations use the feature h before the projector; others use intermediate or patch-level transformer features. A low pretraining loss measured on z is not itself evidence that the feature h has the same geometry or usefulness. The projector can change the geometry substantially because it is a learned, generally nonlinear map. The pretraining loss directly constrains z, not every pairwise relation in h. Claims about representation quality should state the tensor and evaluation protocol rather than infer quality from training loss alone.

3.5.3. A global vector or a grid of local features?

A global vector is compact and convenient for classification, retrieval and simple interfaces. A grid of patch features preserves approximate locations and correspondences. A temporal sequence preserves history only if its aggregation does not erase the changes needed by prediction. These

forms are related but not interchangeable.

For patch features $h_1, \ldots, h_P$, simple average pooling forms

$$h_{\text{global}} = \frac{1}{P} \sum_{i=1}^{P} h_i.$$

This compresses the grid into one vector and removes explicit patch indexing from the output. Distinct spatial arrangements can therefore produce the same pooled vector. Retaining the grid preserves a feature for each patch and its token position, although the feature values themselves are not guaranteed to encode accurate geometry. The choice is a representational trade-off: a vector is compact, while a grid makes local correspondences accessible to later components. Global pooling is appropriate when spatial arrangement is irrelevant, but can be harmful when the target depends on location or correspondence between regions. A temporal representation also has to retain motion-related information when it changes the target future.

The pooling equation also gives a simple source of aliasing. Consider two feature grids whose patch vectors are identical as a multiset but appear in different spatial orders. Average pooling gives the same sum, hence the same global vector, even though the arrangements differ. If the target depends on the arrangement, the pooled representation cannot be sufficient for that target. Positional information elsewhere in the encoder can reduce such collisions, but the pooling operation itself does not preserve the index of each patch.

3.5.4. Compression and predictive sufficiency

A useful representation retains enough information for the relevant prediction while suppressing nuisance variation. There is no universally optimal latent dimension for all tasks. In some settings a small state may be sufficient; in others, spatial features, temporal context or multiple latent components may be necessary.

Here, **compact** means constrained and usable by downstream prediction, not necessarily information-theoretically minimal. The bottleneck should be selected and tested against the downstream task. A low-dimensional representation that aliases important states is not a success; a larger representation is not automatically better if it mostly stores nuisance detail.

3.6. Compact state representations

An encoder can provide a vector, spatial tokens or temporal context

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Aliasing is a failure of predictive sufficiency

Artwork placeholder: Figure 3.28. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F28_Dadwal.md.

A low self-supervised loss establishes only that the representation satisfies its training objective. It does not establish that the representation is useful for the quantity a downstream model must predict. The distinction is mathematical: each pretraining objective constrains a selected relation, whereas predictive sufficiency concerns the full conditional distribution of a specified target.

A linear probe tests whether a selected property is accessible through a linear map:

$$\hat{y} = Wz + b.$$

A poor score does not prove the property is absent; the representation may encode it nonlinearly. A high score establishes linear accessibility under that probe and data split, not sufficiency for every possible target. Probe results should identify the feature tensor, normalization, readout family, training split and held-out evaluation set. Otherwise, a reported difference can reflect feature selection, readout capacity or data leakage rather than the representation itself.

Other diagnostics measure other things. Per-coordinate variance can detect dimensions that are constant across a batch. The covariance spectrum can reveal whether most feature variation lies in only a few directions. Nearest-neighbor retrieval measures local geometry under a selected distance. Reconstruction evaluates a particular target and decoder. Contrastive retrieval evaluates ranking among a sampled candidate set. These results are not interchangeable: none alone demonstrates that the representation preserves every distinction required to predict the future.

A useful formal definition is **predictive sufficiency**. Let X denote the available observation or observation history, $Z = f(X)$ the representation, and Y the future target. The representation is sufficient for Y if

$$P(Y \mid X) = P(Y \mid Z).$$

Equivalently, $Y \perp X \mid Z$: after conditioning on Z, the full input provides no additional information about Y. This definition is relative to the target distribution. A representation can be sufficient for a coarse category while not being sufficient for a precise location, or sufficient for a one-step target while losing information required for a longer horizon. There is no target-independent scalar score that certifies sufficiency in all these settings.

The failure caused by aliasing follows immediately. Suppose two inputs map to the same representation,

$$f(x_1) = f(x_2) = z, \qquad P(Y \mid x_1) \neq P(Y \mid x_2).$$

A predictor that receives only z must produce the same output distribution for both inputs. It cannot equal both distinct conditional distributions. Increasing the capacity of the downstream predictor cannot restore information removed by the encoder.

Consider two short video histories that end with an object at the same image location. In one history it has been moving left; in the other it has been moving right. If the representation discards the temporal difference and maps both histories to the same z, their next-frame distributions may differ even though the last frames are identical. The example does not imply that velocity must always be explicitly represented as a coordinate. It shows that distinctions that change the target distribution cannot be merged without loss relative to that target.

The practical evaluation should therefore state the claim being tested. A probe supports a claim about accessibility of specified variables. A nearest-neighbor score supports a claim about local feature geometry under a specified metric. A reconstruction metric supports a claim about recovery of a chosen target. A predictive-sufficiency claim is stronger and requires evidence tied to the target distribution and observation history of interest.

A useful audit begins by naming the prediction target, the observations available at inference, and the horizon over which the claim is made. It then identifies the exact tensor exported by the encoder, freezes that encoder for representation probes, and reports the readout, normalization, data split and metric. The evaluation should include at least one measurement tied directly to the intended target, rather than relying only on the pretraining loss or an unrelated benchmark. When temporal or action-conditioned predictions matter, the evaluation must include histories and action conditions relevant to that setting; a static-image probe cannot answer those questions.

The readout also sets a limit on the conclusion. If a linear probe fails but a nonlinear predictor succeeds, the information may be present but not linearly accessible. If both fail, the experiment still cannot prove that the information is absent under every possible decoder; it shows that the tested readouts did not recover it under the given data and optimization budget. Supervised labels used only for this evaluation should be described as evaluation targets, not as part of self-supervised pretraining. These distinctions keep a diagnostic from being promoted into a stronger claim than its design supports.

A training loss and a probe answer different questions

Artwork placeholder: Figure 3.29. Artwork is intentionally absent from this branch. Use the full production brief in images/CH03_F22_Dadwal.md.

3.7. Summary

Self-supervised learning constructs a target relationship from data. Contrastive methods such as

SimCLR and MoCo rank positives against sampled candidates; CPC applies that idea to future latents. BYOL and SimSiam predict detached view targets but differ in how their targets change. DINO matches teacher distributions, Barlow Twins constrains cross-view correlations, and VICReg separates agreement, variance and covariance penalties. SIGReg and VISReg constrain aspects of feature-distribution shape, while clustering methods create pseudo-targets from current features.

Masked methods differ in whether they predict pixels, discrete codes or teacher features; JEPAs instead predict learned target representations from context. Across these families, the target and loss specify which relationships training rewards, while the encoder architecture and output bottleneck determine where those constraints act. None of these ingredients, nor a low pretraining loss alone, establishes predictive sufficiency. That claim must be evaluated for a named target, input history and prediction horizon.

3.8. Exercises

1. For an anchor with one positive and $K-1$ negatives, derive the softmax-cross-entropy gradient with respect to each scaled similarity logit. Explain why a high-probability negative receives a larger gradient than a low-probability negative.
2. A frozen encoder is excellent at image retrieval but weak at predicting motion. Give two plausible reasons for the mismatch, then design a target-specific evaluation that distinguishes missing temporal information from information that is present but difficult for a linear readout to access.
3. State the parameter-update difference between BYOL and SimSiam. In each method, which pathway receives gradients, and which mechanism changes the target?
4. Write the Barlow Twins objective in terms of the cross-correlation matrix. Explain the role of its diagonal and off-diagonal terms, then contrast it with VICReg's three terms.
5. Compare the targets used by MAE, BEiT and iBOT/data2vec. For each method, state what is predicted and what the loss measures.
6. Give two observation histories x_1, x_2 that map to the same representation but have different conditional future distributions. Explain why no downstream predictor that receives only that representation can predict both distributions correctly.
7. Decompose DINO's teacher-student cross-entropy into target entropy and KL divergence. Explain what happens if the teacher returns the same distribution for every image.
8. For a batch whose second VICReg coordinate is constant but whose first coordinate varies, determine which term penalizes the second coordinate and why the covariance penalty alone would not be enough.

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